

Supplementary Table S1

Supplementary Table S1 — Reference cut-points for rule- based morphological categorization and Standardized Caliber Uniformity Score

Cut-points for the Complexity Score and the Standardized Caliber Uniformity Score (NV Diameter CV and dilated vessel fraction) are shown by imaging platform and field of view. Thresholds are stratum-specific and intended for within-platform classification and reporting; they do not imply biological equivalence across devices. For Optovue Solix (6×6 mm), the Complexity Score percentile reference was derived from n = 34 eyes; the primary analysis cohort in Tables 2–3 included n = 33 eyes in this stratum.

A. Complexity Score percentile thresholds

The P30 threshold is obtained by linear interpolation between P25 and P40 and is not tabulated as a stored percentile cut-point, consistent with the morphological classification rule.

Stratum (device)	n	P10 (Dead tree <)	P30 (Glomerular ≥; interpolated)	P40 (Seafan ≥ with SEAFAN trunk)	P65 (Medusa ≥ with MEDUSA trunk)
Zeiss PlexElite (6×6 mm)	49	24.29	38.36	43.21	55.00
Zeiss CIRRUS HD-OCT with AngioPlex (3×3 mm)	30	44.77	67.93	69.36	72.85
Optovue Solix (6×6 mm)	34	34.20	41.21	44.93	53.77

Operational priority order: Dead tree → Medusa → Seafan → Glomerular → Tree in bud (remainder). Trunk pattern (MEDUSA / SEAFAN / INTERMEDIATE) is derived from spatial organization of large-caliber segments.

B. Complexity Score percentiles

Stratum	P10	P25	P40	P50	P60	P65	P75	P90
Zeiss PlexElite (6×6 mm)	24.29	35.93	43.21	48.81	53.32	55.00	57.57	71.67
Zeiss CIRRUS HD-OCT with AngioPlex (3×3 mm)	44.77	67.21	69.36	71.03	72.29	72.85	74.31	77.74
Optovue Solix (6×6 mm)	34.20	39.34	44.93	48.71	51.06	53.77	55.65	65.22

C. Standardized Caliber Uniformity Score

NV Diameter CV and Dilated vessel fraction cuts

Score = clip($0.75 \cdot U(-\text{NV Diameter CV}) + 0.25 \cdot U(-\text{Dilated vessel \%})$, 0, 100), where U maps stratum min/median/max of the negated feature to 0/50/100.

Stratum (device)	n	NV-CV min	NV-CV median	NV-CV max	Dilated% min	Dilated% median
Zeiss PlexElite (6×6 mm)	49	36.5120	38.4867	40.9939	0.076141	0.130156
Zeiss CIRRUS HD-OCT with AngioPlex (3×3 mm)	30	43.0772	47.2952	79.1192	0.046519	0.091894
Optovue Solix (6×6 mm)	33	37.0858	41.3097	48.8622	0.041908	0.076640